

Lending Club Credit Risk & Loan Default Analysis

Exploratory Data Analysis (EDA) Summary

Dataset: All Lending Club loan data. The analysis used 2,260,701 loan records and a selected set of 27 columns. For default-rate analysis, only resolved loans were used: Fully Paid and Charged Off loans.

1. Data Preparation & Quality Checks

- **Dataset size:** 2,260,701 rows and 27 selected columns.
- **Duplicate rows:** 0. **Duplicate IDs:** 0.
- **Main missing values:** emp_length 146,940 (6.50%); revol_util 1,835 (0.081%); dti 1,744 (0.077%). Several other columns had only 62, 37, or 33 missing values.
- **Loan status:** Fully Paid 1,076,751; Current 878,317; Charged Off 268,559; Late (31-120 days) 21,467; In Grace Period 8,436; Late (16-30 days) 4,349. A few records had other statuses or missing status.
- **Why resolved loans?** Current and late loans have not reached a final outcome. Fully Paid and Charged Off give a clear completed outcome, so they were used to calculate charge-off rates.

2. Target Variable

A default_flag was created: Charged Off = 1 and Fully Paid = 0. The resolved-loan dataset contained 1,345,310 loans: 1,076,751 Fully Paid and 268,559 Charged Off. Overall charge-off rate was 19.96%.

Important: Charge-off is the outcome used in this project. The analysis shows association, not proof that a factor causes default.

3. EDA Results

Loan Grade

Charge-off rate increased strongly from Grade A (6.04%) to B (13.39%), C (22.44%), D (30.38%), E (38.48%), F (45.20%), and G (49.93%).

Interest Rate

5–10%: 8.16%; 10–15%: 18.58%; 15–20%: 29.50%; 20–25%: 38.39%; 25–30%: 47.54%; 30–35%: 49.02%.

Higher-rate bands had higher charge-off rates. Interest rate is closely related to grade, so it should not be treated as an independent cause.

Loan Term

36-month loans had a 15.99% charge-off rate, compared with 32.45% for 60-month loans.

Annual Income

Mean income was \$76,247 and median income was \$65,000. The distribution was right-skewed. Charge-off rates generally decreased as income increased: \$1–25K 24.47%, \$25–50K 22.70%, \$50–75K 20.48%, \$75–100K 18.25%, \$100–150K 16.18%, \$150–250K 14.94%, and >\$250K 13.00%.

DTI

DTI had mean 18.28 and median 17.61. Invalid/out-of-range values were present, including negative values, 999, and values above 100. Using valid 0–100 values, charge-off rates rose from 14.88% for DTI 0–10 to 30.86% for DTI >40.

FICO

FICO midpoint averaged 698.19, with median 692 and range 627–847. Charge-off rates generally fell as FICO increased: 660–679 25.28%; 680–699 21.44%; 700–719 17.28%; 720–739 13.80%; 740–759 11.15%; 760–779 9.53%; 780+ 7.10%. The 620–659 group had only 2 loans, so it is not meaningful for comparison.

Home Ownership

RENT 23.22%; OWN 20.62%; MORTGAGE 17.21%. Small categories such as ANY, OTHER, and NONE had small sample sizes.

Verification Status

Verified 23.85%; Source Verified 20.95%; Not Verified 14.67%. This is an observed association, not evidence that verification itself causes default.

Loan Purpose

Small business had 29.71%; moving 23.35%; house 21.88%; medical 21.78%; debt consolidation 21.15%; other 21.04%; vacation 19.17%; major purchase 18.60%; home improvement 17.72%; educational 17.18%; credit card 16.93%; car 14.68%; wedding 12.16%. Debt consolidation was by far the largest volume category.

State

Examples of higher observed rates included MS 26.08%, NE 25.18%, AR 24.09%, AL 23.63%, OK 23.48%, and LA 23.18%. California had the largest charge-off count (38,540), while its rate was 19.61%. Geographic differences are associations, not causal effects.

Employment Length

Observed charge-off rates varied across employment-length groups. The missing employment-length group was large enough to matter (78,511 resolved loans, 26.91%).

Open Accounts

The number of open accounts was examined as a credit-history characteristic. Its relationship with charge-off risk was relatively modest compared with stronger variables such as grade, interest rate, DTI, and FICO.

Total Credit Accounts

0–9 accounts: 21.35%; 10–19: 20.61%; 20–29: 19.62%; 30–39: 19.21%; 40–49: 19.51%; 50+: 20.15%. The pattern was not strongly linear.

4. Which Factors Were Most Strongly Associated With Default?

Based on the size and consistency of the observed differences in this EDA, the clearest patterns were seen in **loan grade, interest rate, FICO score, DTI, and loan term**. Grade showed the largest clean step-by-step difference: 6.04% in Grade A versus 49.93% in Grade G. Interest rate also showed a strong gradient, while higher FICO and lower DTI were associated with lower charge-off rates. The 60-month term had roughly twice the charge-off rate of the 36-month term (32.45% vs 15.99%). Income showed a weaker but still visible pattern. Home ownership, verification, purpose, geography, employment length, and account counts showed more moderate or less consistent differences.

5. What is FICO?

FICO is a credit score used to summarize a borrower's credit history and estimate how likely the borrower is to repay credit. In simple terms, a higher FICO score generally means a stronger credit history and lower observed credit risk. In this dataset, FICO was represented by a low and high range, so the analysis used the midpoint: $(\text{fico_range_low} + \text{fico_range_high}) / 2$. The average midpoint was 698.19 and the median was 692.

In the EDA, the relationship was clear: higher FICO bands generally had lower charge-off rates. For example, 660–679 had a 25.28% charge-off rate, while 780+ had 7.10%.

6. Final Business Takeaway

The Lending Club portfolio shows clear differences in charge-off rates across borrower credit quality, debt burden, loan pricing, and loan structure. Lower grades, higher interest-rate bands, higher DTI, lower FICO scores, and longer loan terms were associated with higher observed charge-off rates. These factors can help Lending Club segment borrowers and understand portfolio risk, but this EDA alone does not prove causation or create a predictive credit

model.

7. Important Analysis Notes

- Current and late loans were not treated as defaults because their final outcome is not yet known.
- DTI values outside the sensible 0–100 range were excluded from the DTI band analysis.
- Small groups were interpreted cautiously because a small number of loans can make a rate unstable.
- total_pymnt and recoveries were treated as post-loan outcome variables and were not used as borrower risk predictors.
- This is exploratory analysis. Association should not be interpreted as causation.